

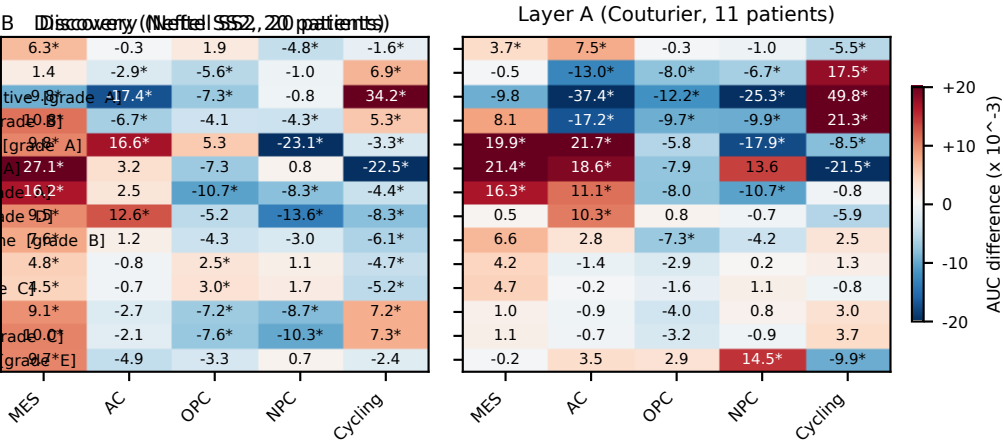
A Two independent single-cell and the preregistered Layer B coverage gate

Layer A (Discovery): Neftel SS2 adult GBM, 20 patients / 4,916 malignant cells / 5 malignant states (8 pediatric patients, sensitivity only)

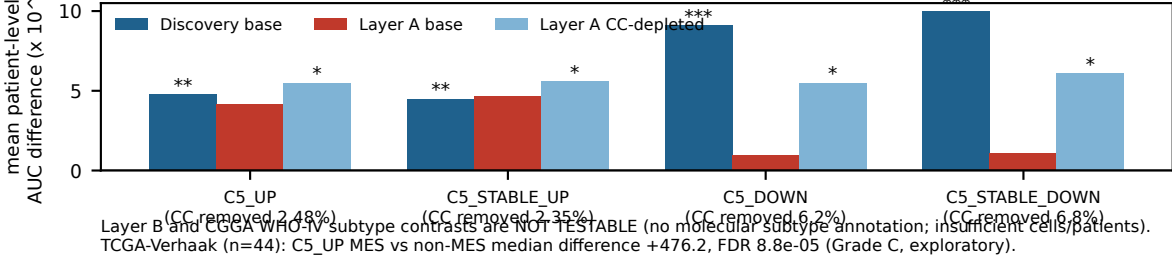
Layer A (Replication): Couturier GBM tissue, 12 samples / 11 independent patients / 18,252 malignant cells; state tests on 8,505 core-state-annotated cells

Layer B (CNV-defined candidate cells): 7 patients; infercnvpy; per-state patient-level test required by preregistration

Layer B per-state testability: MES 11 cells/3 patients vs min 3/2; AC 4 cells/2 patients vs min 3/2; OPC 0 cells/0 patients vs min 3/2; NPC 0 cells/0 patients vs min 3/2; Cycling 12 cells/3 patients vs min 3/2 -> NOT TESTABLE



C C5 is directionally concordant but incompletely replicated (Grade C; base Layer A MES n.s., FDR 0.0563)



Layer B and CGA WHO IV subtype contrasts are NOT TESTABLE (no molecular subtype annotation; insufficient cells/patients). TCGA-Verhaak (n=44): C5_UP MES vs non-MES median difference +476.2, FDR 8.8e-05 (Grade C, exploratory).

D Kept kept negative and sensitivity gate entered into the first layer B mapping

Retained negative / inconsistent findings (not removed from the figures):

HSF1 heat-shock Discovery / MES: delta +9.7e-3, FDR 0.00196

HSF1 heat-shock Layer A / MES: delta -0.2e-3, FDR 1 (reversed direction)

HSF1 heat-shock Layer A / NPC: delta +14.5e-3, FDR 0.00465

Grade E: HSF1 heat-shock module is cohort-dependent and not replicated; it is retained as a heterogeneity finding.

N6: Layer B (CNV-defined cells) fails the preregistered coverage gate: per-state patient-level tests NOT TESTABLE.

N7: C5 base Layer A MES contrast not significant (FDR 0.0563); reported as directionally concordant, not replicated.

N9: C4_DOWN expanded-state mapping is attenuated (FDR 0.594); the Cycling anchor is retained with this sensitivity.

N10: Shared anti-proliferative expanded Cycling mapping is not significant (FDR 0.249); primary anchor mapping unchanged.

N3: Autophagy-lysosome (MES) is not significant in human bulk (TCGA n=44 MES P = 0.334); kept as Grade-B sensitivity only.

Layer B: no figure element is derived from Layer B; coverage insufficiency is shown explicitly.